Shamin Chokshi

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EDUCATION

Northeastern University Boston, MA
Master of Science in Data Architecture and Management August 2024
Vellore Institute of Technology Vellore, India
Bachelor of Technology in Computer Science and Engineering. May 2022

SKILLS

- **Languages: Python** [PyTorch, TensorFlow, Keras, Pandas], SQL, C, C++, JavaScript (JS)
- **Tools: Azure AI,** Azure Computer vision, **Vector DB**, AWS Bedrock, **OpenAI**, Databricks, Snowflake, Azure Data Factory, **Anthropic Claude, Langchain, Langraph, Google ADK**
- **Databases:** MySQL, MS SQL Server, Oracle, MongoDB, PostgreSQL, Cassandra, SQLite, DynamoDB
- **Methods:** Generative AI, **AI Agents, Multiagent Architecture**, **MCP**, ML Model Building, **Prompt Engineering, RLHF, RAG, Multimodality**, Forecasting, LLMOps, Model Deployment, Data Analytics, NLP, Retrieval, A/B Testing, GPU, **CI/CD Pipeline**

WORK EXPERIENCE

BRIGHT HORIZONS | *Generative AI Data Scientist* Boston, MA | October 2024 - Present
- Developed a Retrieval-Augmented Generation (RAG)-based AI Agentic Chatbot, offering personalized counseling on student loans and tuition reimbursements, reducing support requests and reducing operational costs by 9%
- Built a multi-agent chatbot that automatically retrieves and visualizes client data from complex reports via natural language prompts, reducing manual lookup time by 90%
- Developed an AI agent using LangGraph and Azure OpenAI to detect alterations and tampering in grade sheets, identifying fraudulent cases and preventing $10M in erroneous tuition reimbursements
- Forecasted unique EdAssist users with 96% accuracy by building a SARIMA-based time series model, enabling data-driven strategic growth projections and indirect revenue forecasting to support more accurate budgeting
- Implemented and deployed a scalable course recommendation engine using Python and Azure ML, boosting employee course and certification registrations by 13%
- Built an end-to-end MLOps pipeline (CI/CD with Jenkins/GitHub Actions) for model training, deployment, and monitoring, integrating a robust evaluation framework (precision, recall, F1-score) to ensure reproducibility, performance tracking, and seamless version control

BRIGHT HORIZONS | *Business Data Analyst Intern* Boston, MA | June 2023 - December 2023
- Implemented a **multimodal hybrid search** chatbot using LangChain, AWS and pinecone voice and text to solve customer queries, significantly reducing manual labor and operational costs by 30%
- Orchestrated KPIs and metrics by extracting data from an EDW using SQL, ETL, and business intelligence tools to track user behavior and web app feature performance, optimizing performance assessment and increasing feature refinement accuracy by 40%
- Leveraged Business Intelligence tools (Power BI, Tableau) to create interactive dashboards and visualizations, enabling real-time data-driven decision-making and improving performance monitoring efficiency by 35% for key stakeholders

SOHO DRAGON SOLUTIONS | *Software Data Engineer (Finance)* New York, NY | January 2022 - August 2022
- Performed data analysis and modeling to deliver insights for optimization of investment portfolios and risk management processes on a dataset of 650,000 financial transactions and structured financial records
- Engineered a framework using network analysis tools such as DAGs, predictive analysis, and community detection to model financial transaction pathways across various financial institutions to visualize investment narratives across 10,000 client profiles
- Designed and optimized data pipelines and workflows for data collection, preprocessing, and analysis for over 1 million financial data points, enhancing data quality and analysis efficiency

SOHO DRAGON SOLUTIONS | *Software Data Engineer (Healthcare)* New York, NY | Jun 2021 – July 2021
- Conducted feature engineering and dimensionality reduction on 10,000+ EHR datasets using Python and SQL, implementing advanced preprocessing techniques that enhanced machine learning model performance by 25%
- Deployed production-ready machine learning pipeline for patient readmission prediction using scikit-learn and TensorFlow, incorporating real-time model inference and A/B testing to reduce readmission rates by 11%
- Executed predictive analytics and clustering algorithms on patient treatment datasets, developing automated ML workflows and ensemble models that identified high-risk patient segments and improved recovery prediction accuracy by 15%

RELEVANT PROJECTS

HANDMADE AI (Northeastern GenAI Hackathon winner) March 2025 – April 2025
- Built an AI-powered image generation pipeline using FastAPI and Streamlit, applying KMeans color segmentation, Canny edge detection, Gaussian blur, bilateral filtering, and computer vision (OpenCV) to transform photos into numbered paint-by-numbers stencils in under 30 seconds — replacing $15–$30 per-image services with a free, open-source alternative..

TWEAK AI May 2025 – October 2025
- Architected a multi-agent autonomous AI pipeline using LangGraph orchestration with 4 specialized GPT-4 agents (keyword extraction, experience refinement, project optimization, skills tailoring), reducing resume tailoring time from 45+ min to under 60 seconds per application.
- Deployed as a Chrome extension with LinkedIn job auto-detection, offering students a $5/month alternative to $25–$40/month competitors like Jobscan and Teal, cutting job application costs by 80%